API Reference Documentation

Version 1.0 • Node.js + Express • Zoho CRM + MSSQL
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Base URL	`http://<rdp-server>:3000/api/v1`
Auth Header	`X-API-Key: <your_api_key>`
Content-Type	`application/json`
Rate Limit	`100 req/min (standard) • 10 req/min (sync/import)`

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1. Overview & Authentication

sandbox-api is a production-grade REST API server that runs on the RDP server. It connects to Zoho CRM to fetch records and stores them locally as JSON snapshots and/or into an MSSQL database on the same RDP instance.

Authentication

Every request to /api/v1/* must include the X-API-Key header matching the value of API_KEY in .env. Requests without a valid key receive a **401 Unauthorized** response.

```
GET /api/v1/crm/records?module=Leads HTTP/1.1
Host: :3000
X-API-Key: your_super_secret_api_key_here
Content-Type: application/json
```

---

2. Error Codes

All error responses follow a consistent envelope:

```
{
"success": false,
"requestId": "a1b2c3d4-...",
"error": {
"code": "VALIDATION_ERROR",
"message": "module is required"
}
}
```

HTTP Status	Error Code	Meaning
400	VALIDATION_ERROR	Request body/query failed Joi schema validation
401	UNAUTHORIZED	Missing or invalid X-API-Key header
404	NOT_FOUND	Record or stored file does not exist
429	RATE_LIMIT_EXCEEDED	Too many requests — back off and retry
500	INTERNAL_ERROR	Unexpected server error (check error logs)
503	SERVICE_UNAVAILABLE	SQL Server unreachable on RDP

3. Health Check

GET	/health	Server liveness probe

Public endpoint — no authentication required. Returns server status, version, and environment.

Response Fields

Field	Type	Description
status	string	"ok" when server is running
version	string	API version from env (e.g. "v1")
env	string	NODE_ENV value
timestamp	string	ISO 8601 timestamp of the response

Sample Response

```
{
"status": "ok",
"version": "v1",
"env": "production",
"timestamp": "2025-03-09T10:30:00.000Z"
}
```

---

4. CRM Endpoints

All CRM routes are prefixed with /api/v1/crm. They interact with Zoho CRM in real time or read/write from the RDP local storage.

4.1 Fetch Records (Live)

GET	/crm/records	Live paginated fetch from Zoho CRM

Fetches records directly from Zoho CRM using a valid OAuth2 access token. The token is cached in-memory and auto-refreshed on expiry.

Parameters

Parameter	Type	Required	Description
module	string	Yes	Leads | Contacts | Accounts | Deals | Tasks
page	integer	No	Page number. Default: 1
per_page	integer	No	Records per page. Default: 50, Max: 200

fields	string	No	Comma-separated field names to return
sort_by	string	No	Field name to sort by
sort_order	string	No	asc or desc. Default: desc

Sample Request

```
GET /api/v1/crm/records?module=Leads&page;=1&per;_page=50
X-API-Key: your_key
```

Response Fields

Field	Type	Description
success	boolean	true on success
module	string	Module name echoed back
page	integer	Current page
per_page	integer	Page size
info	object	Zoho pagination info (more_records, count, etc.)
count	integer	Number of records in this page
data	array	Array of CRM record objects

Sample Response

```
{
"success": true,
"module": "Leads",
"page": 1,
"per_page": 50,
"info": { "per_page": 50, "count": 50, "page": 1, "more_records": true },
"count": 50,
"data": [ { "id": "123", "Last_Name": "Smith", ... } ]
}
```

4.2 Fetch Single Record

GET **/crm/records/:module/:id** Fetch one CRM record by ID

Returns a single record from the specified Zoho CRM module by its unique record ID.

Parameters

Parameter	Type	Required	Description
module	string (path)	Yes	CRM module name (e.g. Leads)

id	string (path)	Yes	Zoho record ID

Sample Request

```
GET /api/v1/crm/records/Leads/4876893000001234567
X-API-Key: your_key
```

Response Fields

Field	Type	Description
success	boolean	true on success
module	string	Module name
data	object	The CRM record object

Sample Response

```
{
"success": true,
"module": "Leads",
"data": {
"id": "4876893000001234567",
"Last_Name": "Smith",
"First_Name": "John",
"Email": "john@example.com"
}
}
```

4.3 Sync Module to RDP Storage

POST **/crm/sync** Full sync — fetch all records and save to disk

Auto-paginates through ALL records in the given Zoho module and writes them to storage/crm-data/{Module}/latest.json plus a timestamped snapshot. Rate-limited to 10 requests/min. If data was synced within 30 minutes, returns the existing data unless overwrite=true.

Parameters

Parameter	Type	Required	Description
module	string	Yes	Module to sync: Leads | Contacts | Accounts | Deals
overwrite	boolean	No	Force re-sync even if data is fresh. Default: false

Sample Request

```
POST /api/v1/crm/sync
```

```
X-API-Key: your_key
Content-Type: application/json

{ "module": "Leads", "overwrite": false }
```

Response Fields

Field	Type	Description
success	boolean	true on success
module	string	Synced module name
count	integer	Total records fetched and saved
latestPath	string	Absolute path to latest.json on RDP
snapshotPath	string	Absolute path to the new snapshot file
syncedAt	string	ISO timestamp of the sync

Sample Response

```
{
"success": true,
"message": "Sync complete for Leads",
"module": "Leads",
"count": 1842,
"latestPath": "./storage/crm-data/Leads/latest.json",
"snapshotPath": "./storage/crm-data/Leads/snapshots/2025-03-09T10-30-00.json",
"syncedAt": "2025-03-09T10:30:00.000Z"
}
```

4.4 Read Stored Data

GET **/crm/storage/:module** Read latest.json from RDP storage

Returns the most recently synced data for the module from local RDP disk. Does not call Zoho — reads from storage only.

Parameters

Parameter	Type	Required	Description
module	string (path)	Yes	CRM module name

Sample Request

```
GET /api/v1/crm/storage/Leads
X-API-Key: your_key
```

Response Fields

Field	Type	Description
success	boolean	true on success
module	string	Module name
fetchedAt	string	ISO timestamp of when data was synced
count	integer	Total records stored
records	array	Full array of saved CRM records

Sample Response

```
{
"success": true,
"module": "Leads",
"fetchedAt": "2025-03-09T10:30:00.000Z",
"count": 1842,
"records": [ ... ]
}
```

4.5 List Snapshots

GET **/crm/storage/:module/snapshots** List all snapshots for a module

Returns filenames of all timestamped snapshots on disk, newest first. Snapshots older than 30 days are pruned automatically.

Parameters

Parameter	Type	Required	Description
module	string (path)	Yes	CRM module name

Sample Request

```
GET /api/v1/crm/storage/Leads/snapshots
X-API-Key: your_key
```

Response Fields

Field	Type	Description
success	boolean	true on success
module	string	Module name
count	integer	Number of snapshot files

snapshots	array	List of snapshot filenames (newest first)

Sample Response


```
{
"success": true,
"module": "Leads",
"count": 14,
"snapshots": [
"2025-03-09T10-30-00.json",
"2025-03-08T10-15-00.json",
...
]
}
```


---

5. SQL Endpoints

All SQL routes are prefixed with /api/v1/sql. They connect to the MSSQL Server running on the RDP instance via a connection pool.

5.1 Query Table

GET	/sql/query	SELECT rows from an MSSQL table

Runs a safe parameterised SELECT with optional WHERE, LIMIT, and OFFSET. Uses OFFSET/FETCH NEXT for pagination (SQL Server 2012+).

Parameters

Parameter	Type	Required	Description
table	string	Yes	Table name (alphanumeric only, max 128 chars)
where	object	No	JSON object of column:value equality filters
limit	integer	No	Max rows to return. Default: 100, Max: 1000
offset	integer	No	Rows to skip for pagination. Default: 0

Sample Request

```
GET /api/v1/sql/query?table=crm_leads&limit;=50&offset;=0
X-API-Key: your_key
```

Response Fields

Field	Type	Description
success	boolean	true on success
table	string	Table name echoed back
count	integer	Number of rows returned
data	array	Array of row objects

Sample Response

```
{
"success": true,
"table": "crm_leads",
"count": 50,
"data": [
{ "id": 1, "zoho_id": "123", "last_name": "Smith", ... },
...
]
```

```
}
```

5.2 Bulk Insert

POST **/sql/insert** Bulk INSERT rows into an MSSQL table

Inserts one or more rows into a SQL table using a transaction. All rows are committed atomically — if any row fails, the entire batch is rolled back.

Parameters

Parameter	Type	Required	Description
table	string	Yes	Target table name
data	array|object	Yes	Single object or array of objects to insert

Sample Request

```
POST /api/v1/sql/insert
X-API-Key: your_key
Content-Type: application/json

{
"table": "crm_leads",
"data": [
{ "zoho_id": "123", "last_name": "Smith", "email": "j@x.com" },
{ "zoho_id": "124", "last_name": "Jones", "email": "k@x.com" }
]
}
```

Response Fields

Field	Type	Description
success	boolean	true on success
table	string	Target table
inserted	integer	Number of rows successfully inserted
message	string	Human-readable confirmation message

Sample Response

```
{
"success": true,
"table": "crm_leads",
"inserted": 2,
"message": "Successfully inserted 2 row(s) into crm_leads"
```

```
}
```

5.3 Import CRM Storage into SQL

POST **/sql/import/:module** Import latest CRM JSON file into SQL table

The "full pipeline" endpoint. Reads the latest synced CRM JSON from disk (storage/crm-data/{Module}/latest.json) and bulk-inserts all records into the specified SQL table. Run /crm/sync first to populate the storage.

Parameters

Parameter	Type	Required	Description
module	string (path)	Yes	CRM module name (must have a latest.json)
table	string (body)	Yes	Target SQL table name

Sample Request

```
POST /api/v1/sql/import/Leads
X-API-Key: your_key
Content-Type: application/json

{ "table": "crm_leads" }
```

Response Fields

Field	Type	Description
success	boolean	true on success
module	string	CRM module imported
table	string	Target SQL table
inserted	integer	Number of records inserted
message	string	Confirmation message

Sample Response

```
{
"success": true,
"module": "Leads",
"table": "crm_leads",
"inserted": 1842,
"message": "Imported 1842 records from Leads storage into crm_leads"
}
```

5.4 SQL Health Check

GET	/sql/health	SQL Server connectivity probe

Runs a SELECT 1 against the MSSQL connection pool. Returns 200 if connected, 503 if not.

Sample Request

```
GET /api/v1/sql/health
X-API-Key: your_key
```

Response Fields

Field	Type	Description
success	boolean	true if connected, false if unreachable
service	string	"sql"
status	string	"connected" or "unreachable"
checkedAt	string	ISO timestamp

Sample Response

```
{ "success": true, "service": "sql", "status": "connected",
"checkedAt": "2025-03-09T10:30:00.000Z" }
```

---

6. Logging Reference

All logs are written to the ./logs/ directory on the RDP server using Winston with daily file rotation and gzip compression.

Log File Pattern	Level	Retention	Contents
combined-YYYY-MM-DD.log	All	30 days	Every log statement across all modules
error-YYYY-MM-DD.log	error	30 days	Errors only — SQL failures, Zoho errors, crashes
http-YYYY-MM-DD.log	http	14 days	Morgan HTTP access log — every request/response

Log Entry Format (JSON)

```
{
"timestamp": "2025-03-09 10:30:00",
"level": "info",
"message": "Zoho: fetching records",
"module": "Leads",
"page": 1,
"per_page": 50
}
```

7. Environment Variables

Variable	Required	Default	Description
NODE_ENV	No	development	production | development | test
PORT	No	3000	HTTP port the server listens on
API_KEY	Yes	—	Secret key for X-API-Key auth
RATE_LIMIT_WINDOW_MS	No	60000	Rate limit window in milliseconds
RATE_LIMIT_MAX_REQUESTS	No	100	Max requests per window
ZOHO_CLIENT_ID	Yes	—	Zoho OAuth2 client ID
ZOHO_CLIENT_SECRET	Yes	—	Zoho OAuth2 client secret
ZOHO_REFRESH_TOKEN	Yes	—	Zoho OAuth2 refresh token
ZOHO_REDIRECT_URI	No	zohoapis.com/crm/v2	Zoho redirect URI
SQL_SERVER	Yes	—	MSSQL server hostname or IP
SQL_DATABASE	Yes	—	Database name
SQL_USER	Yes	—	SQL login username
SQL_PASSWORD	Yes	—	SQL login password
SQL_PORT	No	1433	SQL Server port
SQL_ENCRYPT	No	true	Use TLS encryption (true|false)
SQL_TRUST_CERT	No	false	Trust self-signed cert (true|false)
STORAGE_BASE_PATH	No	./storage/crm-data	CRM JSON storage root
LOG_LEVEL	No	info	Winston log level
LOG_DIR	No	./logs	Directory for log files